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Original article

Port logistics service quality and customer satisfaction: Empirical evidence from Vietnam

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ABSTRACT

This paper aims to validate five determinants of service quality and to examine the service quality-customer satisfaction link in the port logistics service industry of a developing and transitional economy. First, the research reviews literature pertinent to service quality and customer satisfaction. Second, it uses both qualitative and quantitative methods through focus group discussion and direct interviews with 212 respondents who are the employees of firms that have been using port logistics service provided by Cat Lai Port, Ho Chi Minh City, Vietnam. Finally, the multivariate analysis is subsequently employed to analyze data obtained from surveys by structural equation modeling (SEM) technique. Findings indicate that port logistics service quality is positively determined by five factors including responsiveness, assurance, reliability, tangibles and empathy. In addition, port logistics service quality exerts positive influence on customer satisfaction. Most noticeably, as embedded in the tangibles component, technological advancements appear to enhance service quality which ultimately satisfies customers in the port logistics service industry.

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1. Introduction

Cat Lai Port is the most strategically important traffic hub of Ho Chi Minh City in particular and Vietnam in general (Nguyen & Tran, 2018). This is the river port where important international trading activities take place, which contributes immensely to Vietnam's economic growth and greater integration into global and regional value chain. Cat Lai Port is managed by the Cat Lai Port Joint Stock Company (Cat Lai Port JSC). As could be seen in Fig. 1, Cat Lai Port is located in the area of Dong Sea of Vietnam (South China Sea), which is the bridge connecting the Pacific and Indian Ocean. In addition, it is neighboured by Strait of Malacca which is one of the primary chokepoints in the main maritime shipping routes.[1] Furthermore, there are six secondary chokepoints surrounding the location of Cat Lai Port. This makes Cat Lai Port become a potential transshipment point of the global shipping networks.

Currently, it has provisioned a wide range of port logistics services, specifically shipping agency, sea freight agency, maritime brokerage, supply of ships, goods tallying, ship towage, at-port ship repair, ship cleaning, loading and unloading, multimodal transport, repairing and building containers and trailers for specialized use, trading and leasing ships, containers and transport equipment of over-sized and over-weighted goods, civil engineering, industry and irrigation building, customs declaration, transporting goods by road and inland waterway, vehicles and handling equipment repair and maintenance. In 2018, revenue of Cat Lai Port increased by 50.14% of which the revenue of logistics service increased by 167.4%.[2]

The port logistics service model of Cat Lai Port consists of three components, namely *port staff*, *port logistics service delivery system*, and *clients*. Information flows between those components are based on *soft* and *hard platform*. Soft platform is the information system of port logistics management and the hard platform is physical infrastructure, machine and equipment for handling shipments at port (see Fig. 2).

In terms of *hard platform*, Cat Lai Port has been invested in a modern system of RTG Crane 6 + 1 for enhancing container handling productivity (ship-to-shore crane) and a fleet truck tractors and

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[1] According to the map of Main Maritime Shipping Routes produced by Dr. Jean-Paul Rodrigue, Dept. of Global Studies & Geography, Hofstra University, https://transportgeography.org/?page_id=2067.

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[2] Annual Report 2018 of Cat Lai Port JSC.

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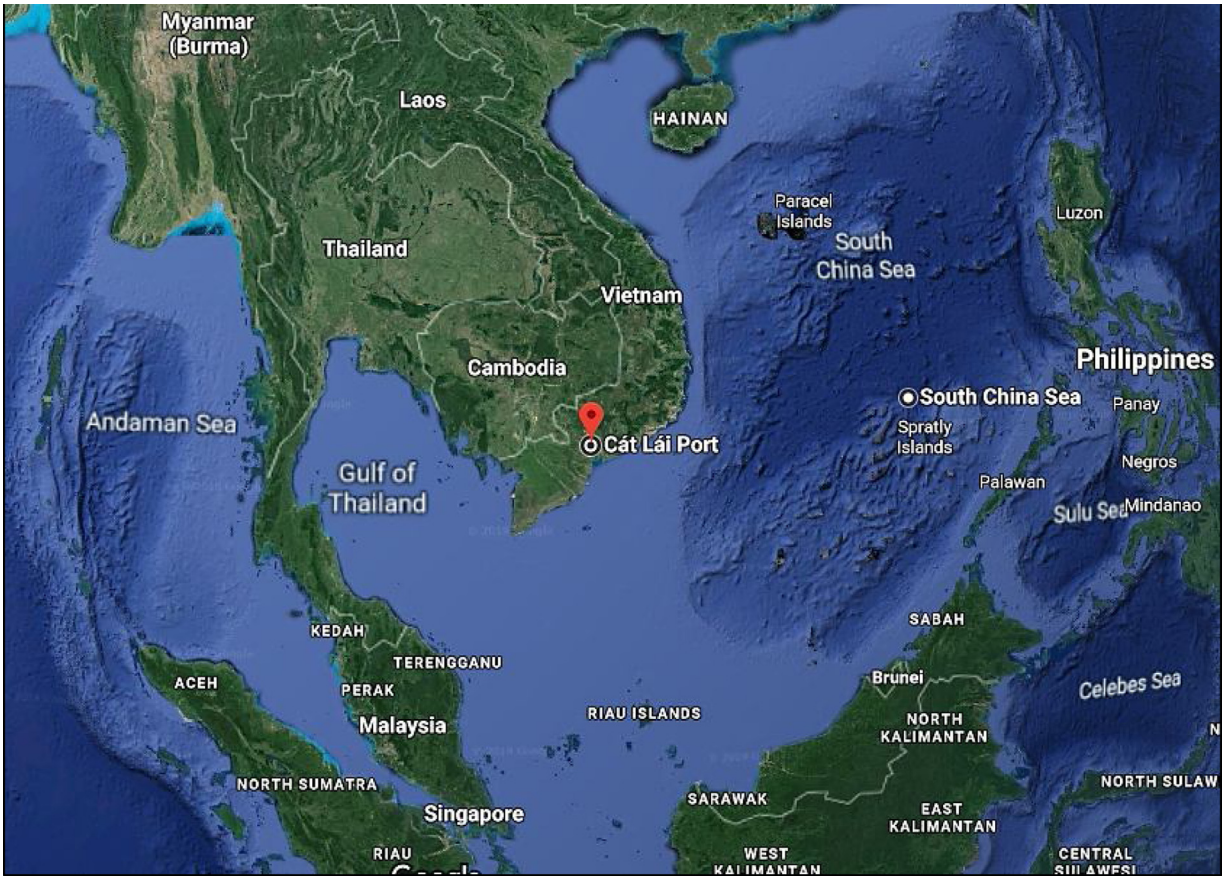

Fig. 1. Location of Cat Lai Port, Ho Chi Minh city, Vietnam.

Source: Satellite photo of Google Map.

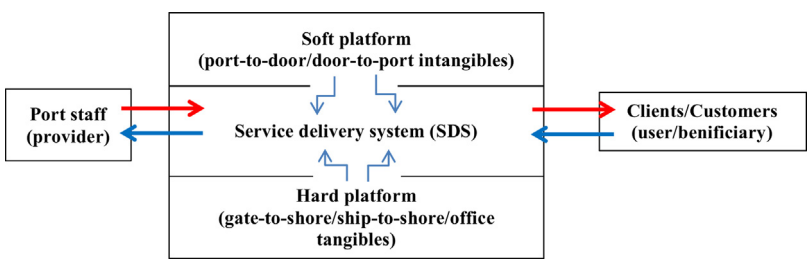

Fig. 2. Port logistics service delivery canvas of Cat Lai Port.

Source: Proposed by the Authors.

semi-trailers for solving internal congestions. In terms of infrastructure, Cat Lai Port has opend Terminal C in addition to terminals A & B. This would strengthen the warehousing capacity of Cat Lai Port, enhance readiness of port infrastructure to serve large-scale corporate clients and provide more space for added-value services provisioning. In terms of *soft platform*, port management information system has been upgraded and the personnel has been extended. In 2013, Cat Lai Port established a new Department of Logistics Management which marked the strategic switch to the port logistics service industry mounting further steps into this fertile competitive market. The red arrows represent the service flow running from port staff to customers throung the SDS which is based on the tangible and intangible platforms. Meanwhile, the blue arrows represent financial and information flows from customers to port staff. Tangible and intangible platforms shape the interactive environment for port staff and customers. Therefore, when using port logistics services, customers interact with and consequently gain satisfaction from those components.

Recently, there are a number of downsides at Cat Lai Port regarding its technological and managerial capacity (Nguyen & Tran, 2018) that the board management of this seaport needs to find a solution to overcome as technology and management have been the key advantages of port logistics providers in the age of digitalization, expanded global economy and efficiency-based competition (Cimino et al., 2017; Kim, 2014; Loh & Thai, 2016; Nguyen, Nguyen, Chang, Chin, & Tongzon, 2016; Nguyen, Nghiem, & Chang, 2018; Yang & Chen, 2016). Many of shipments and containers backlogs are under-handled because their owners do not come to receive the goods. This had led to severe intra-port congestions, regularly-limited availability of warehousing area and poor environmental performance in nearby areas. In addition, container trucks enter and leave the port for loading and unloading of goods are constantly delayed and lined up due to the large number of uncleared goods. The most serious issue is that staff members of Cat Lai Port with the container truck drivers adversely affect the reputation and image of the port.

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Due to the absence of deep water seaports, exports or imports of Vietnam still need to be transshipped at larger ports in Asia such as Hong Kong or Singapore and it takes two or three days more than goods shipped directly from/to a deep-water seaport. In this situation, logistics services at Cat Lai Port need to be guaranteed to achieve the highest standards in order to avoid any delays and stagnations in the transportation of goods. Vietnam is increasingly open with accelerating international trade activities, especially Ho Chi Minh City is the largest and most dynamic economic zone in the country so the volume of goods to be transported at Cat Lai Port has been rising significantly. Therefore, the quality of logistics services at the Cat Lai Port not only need to be continuously maintained at a high level, but also need to be enhanced in the near future. Improving the quality of port logistics services and customer satisfaction at Cat Lai Port is really urgent and must be addressed promptly. Thus, Cat Lai Port is a case which is worth-researching in order to come up with a generalized set of solutions to further gaining customer satisfaction in port logistics service industry. Research objectives of the paper are three-fold. First, the research reviews relevant literature on service quality and customer satisfaction in the context of port logistics service industry. Second, it explores the relationship between the components of the service quality for port logistics service with the level of customer satisfaction. Third, its findings propose solutions to improve factors that are expected to enhance service quality and customer satisfaction in port logistics service industry.

2. Theoretical basement

2.1. Service quality gap model of Parasuraman, Zeithaml, and Berry (1985)

Parasuraman et al. (1985) supposed that when businesses can close the gap between the expectations and perceptions of customers, they could be able to achieve quality of service. Therefore, the service quality model of Parasuraman et al. (1985) is based on the analysis of service quality gaps (see Fig. 3)

Table 1
Dimensions of SERVQUAL model.

	Dimensions	Definition
1	Reliability	Express the ability to deliver timely and consistent services.
2	Responsiveness	Express through eagerness to help customers and quickly resolve problems when errors or unexpected situations occur.
3	Assurance	Express through ability to build customer trust through professionalism, politeness, respect for customers, communication skills and attentive attitude to do the best support for our customers.
4	Empathy	Express through the style of services of employees through attention, care, ability to understand the needs and create feelings of secure and safety for customers.
5	Tangibles	Express through the appearance of facilities, equipment, staff uniforms, items and materials used for performing the services.

Source: Parasuraman et al. (1988).

2.2. The SERVQUAL model of Parasuraman, Zeithaml, and Berry (1988)

The SERVQUAL model, created by Parasuraman et al. in 1988, was used to evaluate customer perceptions of service quality including five dimensions (reliability, responsiveness, assurance, empathy, tangibles). These dimensions are illustrated in Table 1.

2.3. The SERVPERF performance-based evaluation model of Cronin and Taylor (1992)

The research of Cronin and Taylor (1992) studied the concepts and methods of measuring service quality and its relationships with customer satisfaction and their purchase intentions, thus concluding that consumers' perceptions factors are the better predictor of service quality.

The measurement of service quality follows Cronin and Taylor's SERVPERF model is considered a convenient and clear method of measuring service quality based on the performance results of

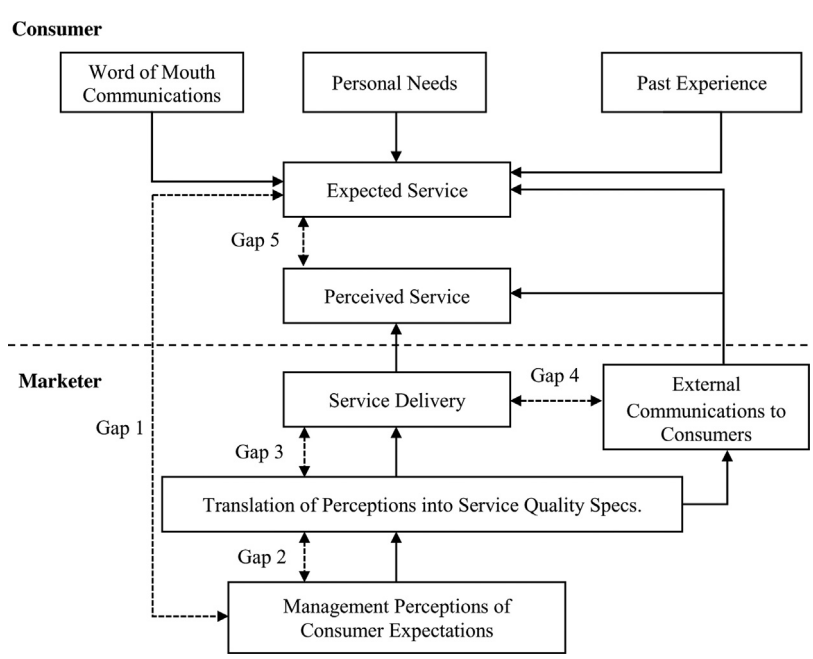

Fig. 3. Model of service quality gap.

Source: *Adapted from* Parasuraman et al., 1985.

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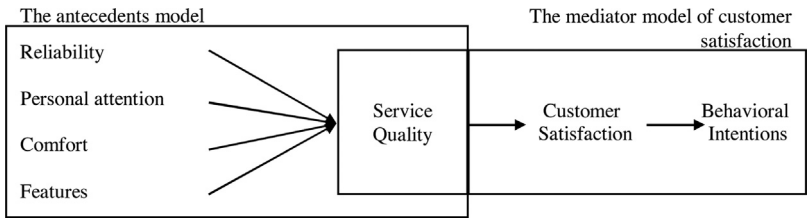

Fig. 4. The antecedents and mediator model

Source: Dabholkar et al., 2000.

service quality. Cronin and Taylor supposed that Parasuraman's SERVQUAL model scale will be easily confused between customer satisfaction and customer attitudes, from that lead to some mistake in measuring quality of service. Cronin and Taylor also indicated that service quality can be defined as "a perception attitude" instead of "performance as expected" and it will better determine the quality of the service based on the actual performance of businesses. According to that, the quality of service is assessed only through the perception of customers without the assessment in the expectations of the customer, which means considered quality of service is exactly the level of perceptions.

2.4. *The antecedents and mediator model of Dabholkar, Shepherd, and Thorpe (2000)*

This model can be considered as a comprehensive model of quality of service, including the consideration of the antecedents, mediators and result to provide a deeper understanding of interrelated concepts of quality of service. This model examines the concepts of service quality with the 4 factors: Reliability, Personal attention, Comfort, Feature that affecting service quality as a component of antecedents and the relationship between service quality with customer satisfaction and behavioral intentions. This model is illustrated in the following Fig. 4.

2.5. *Theoretical backgrounds synthesis*

The theoretical backgrounds synthesis of researched models is illustrated as the Table 2.

Through the analysis of theoretical models, each model has different angles of measuring service quality. In this study, the author selected the SERVQUAL model as the basis for analyzing the current situation and proposing a solution for the quality of Cat Lai Port logistics services. The reasons for choosing SERVQUAL scale for this research will be explained clearly in the next chapter.

Service quality and port service quality (Table 3).

2.6. *Inter-comparison between studies' results*

The following Table 4 will list out some studies that researched on factors of service quality. The table will introduce the sample and the findings results that each factor has affecting on overall service quality/customer satisfaction or not.

From the Table 4, we can see that several studies addressed and confirmed the validity and applicability of 5 components of SERVQUAL model (Empathy, Reliability, Assurance, Responsiveness and Tangibles) in the measuring the quality of service and customer satisfaction in different geographic regions around the world.

Table 2
Theoretical backgrounds synthesis.

	Research model	Dimensions
1	Technical and functional quality model (Gronroos, 1984)	Technical quality; functional quality; image
2	Model of service quality gaps (Parasuraman et al., 1985)	Reliability; responsiveness; competence; access; communication; courtesy; credibility; security; understanding; tangibles
3	SERVQUAL model (Parasuraman et al., 1988)	Reliability, responsiveness; empathy; assurance; tangibles
4	SERPERF model (Cronin & Taylor, 1992)	Reliability; responsiveness; empathy; assurance; tangibles
5	Model of perceived service quality and satisfaction (Spreng & Mackoy, 1996)	Convenience of making appointment; staff's friendliness; staff listen to questions; staff provide accurate information; staff's knowledge; consistent advices, long-range helps; choosing right products helps; interest in personal life; professionalism of offices
6	The antecedents and mediator model (Dabholkar et al., 2000)	Reliability; personal attention; comfort; features

Source: The synthesis of the author.

Table 3
Synthesis of recent literature on port logistics service.

No.	Research model	Characteristics
1	Lopez and Poole (1998)	Efficiency, timeliness, and security to contribute to the quality of port services
2	Brady and Cronin (2001)	The aspects of service quality including "rational quality", "result quality", and "physical environmental quality"
3	Ha (2003)	Identified a group of port service quality factors, including "ready information availability of port-related activities," "port location," "port turnaround time," "facilities available," "port management," "port costs," and "customer convenience."
3	Ugboma, Ibe, and Ogwude (2004)	Five SERVQUAL dimensions were valid
4	Thai (2008)	Service quality consists of the following six dimensions: resources, outcomes, process, management, and image and social responsibility.
5	Cho et al. (2010)	Port service quality comprising "endogenous quality," "exogenous quality," and "relational quality"
6	Yeo, Thai, and Roh (2015)	Port service quality was found to be a five-dimensional construct consisting of items related to resources, outcomes, process, management, and image and social responsibility.

Table 4
The synthesis studies results on factors of quality of service.

Factors	Author (year)	Sample	Findings	Characteristics
Empathy	Parasuraman et al. (1988), Parasuraman, Berry, and Zeithaml (1991)	The U.S.	Positive	This model provides five generic dimensions of service quality, namely tangible, reliability, responsiveness, assurance and empathy.
	Cronin and Taylor (1992)	The U.S.	Positive	This paper in an under-researched Zambian universities context. it examines the influence of each service quality dimension on overall service satisfaction and behavioral intentions in terms of loyalty and positive word of mouth. The findings indicate that service quality performance dimensions (tangibility, reliability, responsiveness, empathy and assurance) are each significantly positively related to overall customer satisfaction which in turn affects behavioral intentions
	Dabholkar et al. (2000)	The U.S.	Positive	This study finds that factors relevant to service quality are better conceived as its antecedents rather than its components and that customer satisfaction strongly mediates the effect of service quality on behavioral intentions.
	Ugboma et al. (2004), Ugboma, Ogwude, Ugboma, and Nnadi (2007)	Nigeria	Positive	five SERVQUAL dimensions were valid
	Pantouvakis and Dimas (2013)	Greece	Not affecting	The purpose of this study is to confirm the applicability of SERVQUAL scale in the shipping corporate sector and explore the relationship between service quality, customer satisfaction, corporate agility and perceived price. Findings were in support to a more parsimonious 3-dimensional one-measurement model arguing thus against the 5-dimensional nature of the SERVQUAL instrument. It has been further supported that agility partially mediates the reliability/responsiveness dimension
		Vietnam	Positive	The purpose of this paper is to define and measure the effects of service quality on customer satisfaction and customer loyalty of four and five star hotels in Ho Chi Minh City, Vietnam. The results show that: (i) customers' loyalty is affected by four dimensions of service quality (reliability, responsiveness, website utility, and tangibles) and customer satisfaction; and (ii) customer satisfaction is affected by five dimensions of service quality (reliability, responsiveness, website utility, tangibles, and sympathy).
	Vo and Nguyen (2012)	Vietnam	Positive	This quantitative research is conducted to identify factors influencing customer satisfaction in Damco Supply Chain Management Service. The research resulted in seven dimensions for customer satisfaction: Tangibles, Reliability, Assurance, and Empathy, Perceived Price, Relationship Service and Relationship
Reliability	Parasuraman et al. (1988), Parasuraman et al. (1991)	The U.S.	Positive	This model provides five generic dimensions of service quality, namely tangible, reliability, responsiveness, assurance and empathy.
	Cronin and Taylor (1992)	The U.S.	Positive	This paper in an under-researched Zambian universities context. it examines the influence of each service quality dimension on overall service satisfaction and behavioral intentions in terms of loyalty and positive word of mouth. The findings indicate that service quality performance dimensions (tangibility, reliability, responsiveness, empathy and assurance) are each significantly positively related to overall customer satisfaction which in turn affects behavioral intentions
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	Vo and Nguyen (2012)	Vietnam	Positive	This quantitative research is conducted to identify factors influencing customer satisfaction in Damco Supply Chain Management Service. The research resulted in seven dimensions for customer satisfaction: Tangibles, Reliability, Assurance, and Empathy, Perceived Price, Relationship Service and Relationship
Assurance	Parasuraman et al. (1988), Parasuraman et al. (1991)	The U.S.	Positive	This model provides five generic dimensions of service quality, namely tangible, reliability, responsiveness, assurance and empathy.
	Cronin and Taylor (1992)	The U.S.	Positive	This paper in an under-researched Zambian universities context. it examines the influence of each service quality dimension on overall service satisfaction and behavioral intentions in terms of loyalty and positive word of mouth. The findings indicate that service quality performance dimensions (tangibility, reliability, responsiveness, empathy and assurance) are each significantly positively related to overall customer satisfaction which in turn affects behavioral intentions
	Ugboma et al. (2004), Ugboma et al. (2007)	Nigeria	Positive	five SERVQUAL dimensions were valid

Table 4 (*Continued*)

Factors	Author (year)	Sample	Findings	Characteristics
	Pantouvakis and Dimas (2013)	Greece	Not affecting	The purpose of this study is to confirm the applicability of SERVQUAL scale in the shipping corporate sector and explore the relationship between service quality, customer satisfaction, corporate agility and perceived price. Findings were in support to a more parsimonious 3-dimensional one-measurement model arguing thus against the 5-dimensional nature of the SERVQUAL instrument. It has been further supported that agility partially mediates the reliability/responsiveness dimension
		Vietnam	Positive	The purpose of this paper is to define and measure the effects of service quality on customer satisfaction and customer loyalty of four and five star hotels in Ho Chi Minh City, Vietnam. The results show that: (i) customers' loyalty is affected by four dimensions of service quality (reliability, responsiveness, website utility, and tangibles) and customer satisfaction; and (ii) customer satisfaction is affected by five dimensions of service quality (reliability, responsiveness, website utility, tangibles, and sympathy).
	Vo and Nguyen (2012)	Vietnam	Positive	This quantitative research is conducted to identify factors influencing customer satisfaction in Damco Supply Chain Management Service. The research resulted in seven dimensions for customer satisfaction: Tangibles, Reliability, Assurance, and Empathy, Perceived Price, Relationship Service and Relationship
Responsiveness	Parasuraman et al. (1988), Parasuraman et al. (1991)	The U.S.	Positive	This model provides five generic dimensions of service quality, namely tangible, reliability, responsiveness, assurance and empathy.
	Cronin and Taylor (1992)	The U.S.	Positive	This paper in an under-researched Zambian universities context. it examines the influence of each service quality dimension on overall service satisfaction and behavioral intentions in terms of loyalty and positive word of mouth. The findings indicate that service quality performance dimensions (tangibility, reliability, responsiveness, empathy and assurance) are each significantly positively related to overall customer satisfaction which in turn affects behavioral intentions
	Ugboma et al. (2004), Ugboma et al. (2007)	Nigeria	Positive	five SERVQUAL dimensions were valid
	Pantouvakis and Dimas (2013)	Greece	Positive	The purpose of this study is to confirm the applicability of SERVQUAL scale in the shipping corporate sector and explore the relationship between service quality, customer satisfaction, corporate agility and perceived price. Findings were in support to a more parsimonious 3-dimensional one-measurement model arguing thus against the 5-dimensional nature of the SERVQUAL instrument. It has been further supported that agility partially mediates the reliability/responsiveness dimension
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	Vo and Nguyen (2012)	Vietnam	Positive	This quantitative research is conducted to identify factors influencing customer satisfaction in Damco Supply Chain Management Service. The research resulted in seven dimensions for customer satisfaction: Tangibles, Reliability, Assurance, and Empathy, Perceived Price, Relationship Service and Relationship
Tangibles	Parasuraman et al. (1988), Parasuraman et al. (1991)	The U.S.	Positive	This model provides five generic dimensions of service quality, namely tangible, reliability, responsiveness, assurance and empathy.
	Cronin and Taylor (1992)	The U.S.	Positive	This paper in an under-researched Zambian universities context. it examines the influence of each service quality dimension on overall service satisfaction and behavioral intentions in terms of loyalty and positive word of mouth. The findings indicate that service quality performance dimensions (tangibility, reliability, responsiveness, empathy and assurance) are each significantly positively related to overall customer satisfaction which in turn affects behavioral intentions
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	Vo & Nguyen (2012)	Vietnam	Positive	This quantitative research is conducted to identify factors influencing customer satisfaction in Damco Supply Chain Management Service. The research resulted in seven dimensions for customer satisfaction: Tangibles, Reliability, Assurance, and Empathy, Perceived Price, Relationship Service and Relationship

Source: Synthesis of the Author (2017).

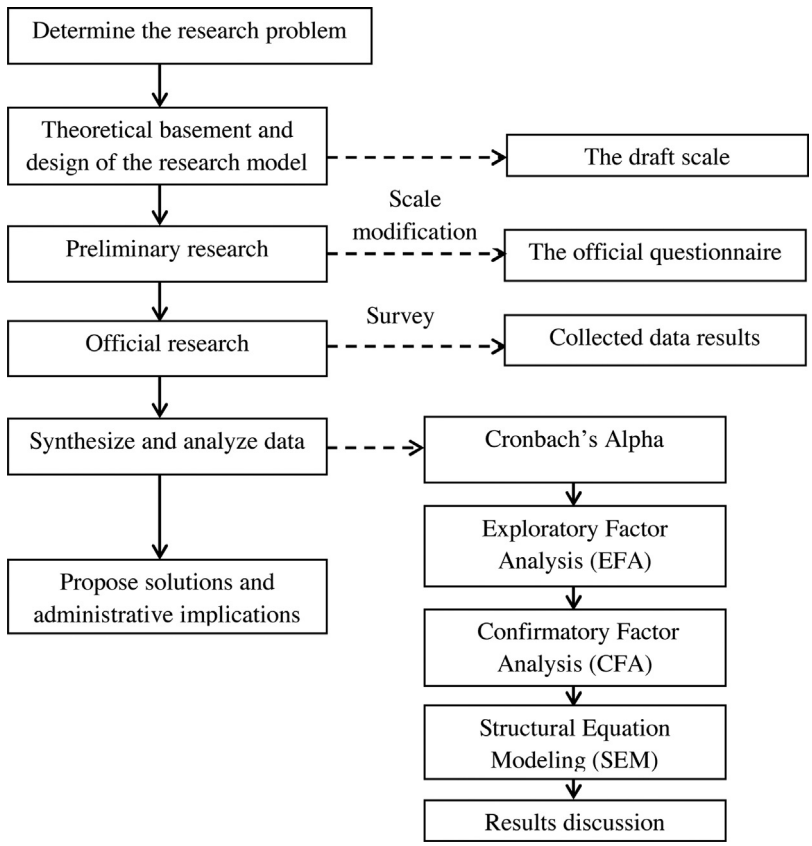

Fig. 5. The research process.

Source: The synthesis of the Author, 2017.

3. Research methodology

3.1. Statistical analysis

We use both qualitative and quantitative approaches through focus group discussion and direct interviews with 212 customers who are the employees of businesses that used port logistics services provided by Cat Lai Port, Ho Chi Minh City, Vietnam. The multivariate analysis method is employed to analyze the data obtained from surveys using structural equation modeling (SEM) technique. SEM technique is used to simultaneously test the causality among reviewed constructs placed in the context of port logistics service industry. The data collected were reviewed during and after collection, uncoded variables were coded, classified, and tabulated. Statistical Package (SPSS) version 20.0 was used for analysis of data. Qualitative data was described.

3.2. Research process

The study was conducted based on a combination of qualitative research and qualitative research. The entire research process will carry out and illustrated in the following Fig. 5.

3.2.1. Preliminary research

Preliminary research is conducted through qualitative methods. With the limit time and resource to complete the study, this research only used the in-depth focus group discussion techniques combine with the synthesis of the author about the related theories, studies, the model and scales that are applied in measure the quality of service, customer satisfaction. In the first step, the author synthesized and adopted the scale from previous studies, translated and then modified the wordings to suitable with the objects for survey to make a draft questionnaire. After that, the author gathered a group of ten people who are working for logistics companies and have at least five years in the field of marine and logistics were invited to the discussion to adjust the observation variables. Focus group which can inform the choice of words or phrases in a questionnaire, the construction of items to measure a given concept, and the development of survey procedures. In addition, focus group allow the author to hear the richly contrasting viewpoints.

All the ideas and suggested modifications in the discussion are used to adjust the questionnaire and form the official scale. The draft questionnaire and the report of the in-depth focus group discussion are in the Appendix of this thesis. This preliminary study was conducted in Ho Chi Minh City in May 2017.

3.2.2. Official research

The official research was conducted by quantitative research. This quantitative research was conducted through direct interviews with 212 customers who are using logistics services at Cat Lai Port by a detailed questionnaire. The field staffs who work for businesses using international seaport logistics services of Cat Lai Port in Ho Chi Minh City, Binh Duong and Dong Nai are the respondents for the questionnaires. They are the representative people of their companies who directly use the logistics services in Cat Lai Port and do the customs clearance procedures for shipments. Therefore, the perceptions and level of satisfaction of these people about Cat Lai Port's logistics services also reflect their companies' perceptions and level of satisfaction. This official research was conducted in Ho Chi Minh City in June 2017.

3.3. Proposed research model

The author proposes the model of research based on the SERVQUAL model to determine the determinants of service quality

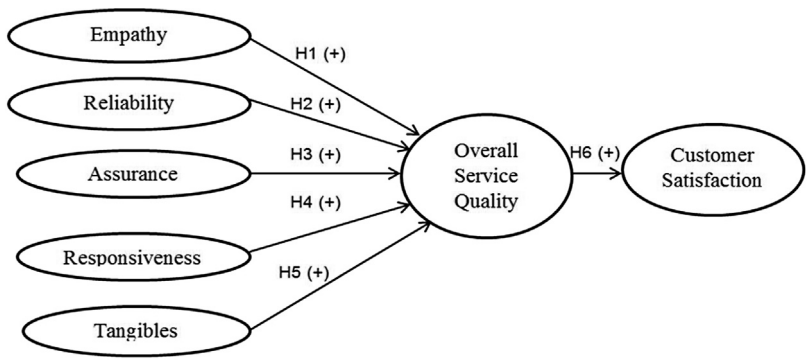

Fig. 6. Proposed research model

and measure customer satisfaction with the logistics services of Cat Lai Port as following Fig. 6.

Research hypotheses:

H_1: Empathy has a *positive* impact on the quality of service (expectation +)

H_2: Reliability has a *positive* impact on quality of service (expectation +)

H_3: Assurance has a *positive* impact on quality of service (expectation +)

H_4: Responsiveness has a *positive* impact on quality of service (expectation +)

H_5: Tangibles has a *positive* impact on quality of service (expectation +)

H_6: Quality of service has a *positive* impact on customer satisfaction (expectation +).

3.4. Sampling method and sample size

3.4.1. Sampling method

The sample was chosen according to the convenience sampling method (non-probability). After carefully considering about time, budget and resource available of this research, the author chose non-probability method with form of convenience sampling.

3.4.2. Sample size

The survey questionnaire in the official study included 26 observational variables for service quality and its components with 4 observational variables for customer satisfaction. So the minimum sample size is: $n \geq 5 * 30 = 150$ samples. Therefore, the minimum sample size required to ensure representativeness and accuracy of the research should be 150 valid survey answers. To ensure high representation and high accuracy for the research, the author collected about 212 samples for the study.

3.5. Scales of measurement

In this content, the scale measurement adopted from previous literature has been presented. Remarkably, the tangible construct has involved technology-based items which aims at further testing the impacts of technology on port logistics service quality and customer satisfaction. See Table 5.

4. Research results

4.1. Sample profile

It could be seen that the number of field staff accounts for a large percentage of the sample which reflects that port logistics service is primarily perceived by front-line executives. In addition, most respondents come from service sector. This is in concert with the fact that in Vietnam forwarding enterprises are prevalent who are the most regular customer of port logistics service.See Table 6.

4.2. Scale reliability results

The results of the verification of the reliability of the service quality scale and customer satisfaction scale are presented in the Appendix and summarized in Table 7.

Reliability Scale: Cronbach's Alpha coefficient = 0.833 is in the required level. The coefficients of the item-total correlation total variables of these variables are accepted (≥0.597). As a result, this scale meets the requirements and the variables of this scale will be processed in the exploratory factor analysis.

Responsiveness Scale: Cronbach's Alpha coefficient = 0.868 is in the required level. The coefficients of the item-total correlation total variables of these variables are accepted (≥0.662). As a result, this scale meets the requirements and the variables of this scale will be processed in the exploratory factor analysis.

Assurance Scale: Cronbach's Alpha coefficient = 0.861 is in the required level. The coefficients of the item-total correlation total variables of these variables are accepted (≥0.643). As a result, this scale meets the requirements and the variables of this scale will be processed in the exploratory factor analysis.

Empathy Scale: Cronbach's Alpha coefficient = 0.875 is in the required level. The coefficients of the item-total correlation total variables of these variables are accepted (≥0.674). As a result, this scale meets the requirements and the variables of this scale will be processed in the exploratory factor analysis.

Tangibles Scale: Cronbach's Alpha coefficient = 0.825 is in the required level. The coefficients of the item-total correlation total variables of these variables are accepted (≥0.597). As a result, this scale meets the requirements and the variables of this scale will be processed in the exploratory factor analysis.

Overall Service Quality Scale: Cronbach's Alpha coefficient = 0.894 is in the required level. The coefficients of the item-total correlation total variables of these variables are accepted (≥0.745). As a result, this scale meets the requirements and the variables of this scale will be processed in the exploratory factor analysis.

Customer Satisfaction Scale: Cronbach's Alpha coefficient = 0.811 is in the required level. The coefficients of the item-total correlation total variables of these variables are accepted (≥0.598). As a result, this scale meets the requirements and the variables of this scale will be processed in the exploratory factor analysis.

Table 5
Scales of measurement.

Reliability	
REL1	Cat Lai Port always performs logistics services which are committed (cargo clearance, loading and unloading containers, inspection, etc.).
REL2	When you encounter services problems (handling, storage,transportation, inspection, fumigation,etc.) at the port, Cat Lai Port proves genuine concern for addressing the problems.
REL3	Logistics services of Cat Lai Port are stable and reliable.
REL4	Cat Lai Port provides the right services at right time as committed.
REL5	In the process of providing logistics services, Cat Lai port does not make any significant mistakes.
Responsiveness	
RES1	Cat Lai Port always informs customers about the time its services are provided to customers.
RES2	The staff membersof Cat Lai Port quickly perform logistics services for you (customs clearance, inspection support...).
RES3	The staff members of Cat Lai Port are always available at your requests when your cargoes face troubles at port.
RES4	The staff members of Cat Lai Port are never too busy to respond to your requests.
Assurance	
ASS1	The behaviors ofstaff members of Cat Lai Port create confidence for you.
ASS2	You feel secure while using the logistics services of Cat Lai Port.
ASS3	The staff members of Cat Lai Port are always polite and courteous to you.
ASS4	The staff members of Cat Lai Port have sufficient expertise to answer your questions about logistics services.
Empathy	
EMP1	The staff members of Cat Lai Port express interest to each customer like you.
EMP2	The staff members of Cat Lai Port know how to care about your logistics services' needs.
EMP3	The staff members of Cat Lai Port understand your special needs related to the transportation of cargoes (warehousing, storage, handling, inspection, etc.).
EMP4	Cat Lai Port pays attention to the issues that you are most concerned about (cut-off time, clearance time, container position, etc.).
EMP5	The working hour of Cat Lai Port makes it convenient to serve your needs.
Tangibles	
TAN1	Cat Lai Port has modern and advanced machinery and equipment.
TAN2	The working offices at Cat Lai Port are spacious, creating trust for you.
TAN3	The staff members of Cat Lai Port are well-dressed and wear neat uniforms.
TAN4	Machinery and equipment of Cat Lai Port (forklifts, warehouses, cranes, piers,etc.) always meet the requirements of providing logistics services.
Overall service quality	
OSQ1	Overall, quality of logistics services provided by Cat Lai Port is superior to those of other ports in Ho Chi Minh City and neighboring areas (Vict, Saigon, Hiep Phuoc, My Thuy, Cai Mep Thi Vai etc.)
OSQ2	There are some types of logistics services provided by Cat Lai Port which have the quality betterthan those of other ports in Ho Chi Minh City and neighboring areas (Vict, Sai Gon, Hiep Phuoc, My Thuy, Cai Mep Thi Vai, etc.)
OSQ3	Overall, the quality of logistics services provided by Cat Lai Port is of high standards in port logistics industry.
OSQ4	Overall, the quality of logistics services provided by Cat Lai Port meets your expectations of service quality.
Customer satisfaction	
SAT1	Overall, you are satisfied when using logistics services of Cat Lai Port.
SAT2	You are satisfied with manners and methods that Cat Lai Port has organized and managed to provide logistics services.
SAT3	You are satisfied with the services provided by and working attitudes of the staff members of Cat Lai Port.
SAT4	You are satisfied with the process of cargo management and cargo customs clearance procedures of Cat Lai Port.

Source: Parasuraman et al. (1988), Parasuraman et al. (1991), Tsou and Liao (2010), Nguyen Van Si & Nguyen Viet Bang (2017).

Table 6
Descriptive analysis of research sample.

No.	Research sample	*N*	Percentage
1	**Job position**		
	Field staff	200	94.3
	Middle–line manager	12	5.7
2	**Types of business**		
	Service enterprise	151	71.2
	Trading enterprise	35	16.5
	Manufacturing enterprise	26	12.3
3	**Work seniority**		
	1–3 years	54	25.5
	Over 3 years	158	74.5
4	**Frequency of service usage**		
	1–3 times/month	45	21.2
	Over 3 times/month	165	78.8

Source: Survey results of the author (2017).

4.3. Exploratory Factor Analysis (EFA) results

4.3.1. Exploratory Factor Analysis (EFA) results for quality of service and its components

The overall service quality scales is group in one factor; its components scales are grouped in five measured factors and the EFA results are used by Varimax rotation method. The Exploratory Factor Analysis (EFA) results are shown in Table 8.

The Bartlett test results in the KMO and Bartlett's test with Sig = 0.000 and the KMO > 0.5 show these 2 factor analyses is appropriate and meets the requirements. At Eigenvalues are greater than 1 (1.848 and 3.034) and the cumulative extraction variance are 67.889%; 75.858% (>50%) which show that factor analyses is appropriate and meet the requirements.

Factor analysis of service quality components has extracted 5 factors from 22 observational variables; factor analysis of service quality has extracted 1 factor from 4 observational variables with Principal components extraction method and Varimax rotation method. The factors loading of all variables are greater than 0.5 so they are appropriate and meet the requirements. Therefore, the author will use these 26 observation variables with 6 factors for further analysis

4.3.2. Exploratory Factor Analysis (EFA) results for customer satisfaction

The customer satisfaction scale is grouped in one measured factors and EFA results are used by Varimax rotation method. The Exploratory Factor Analysis (EFA) results are shown in Table 9.

The Bartlett test results in the KMO and Bartlett's test with Sig = 0.000 and the KMO = 0.755 > 0.5 show that factor analysis is appropriate, meets the requirements. EFA of customer satisfaction is grouped into 1 factor with Principal components extraction

Table 7
Cronbach's Alpha results of factors' scales.

	Scale mean if item deleted	Scale variance if item deleted	Corrected item-total correlation	Cronbach's alpha if item deleted
Reliability Scale (REL): Cronbach's Alpha = 0.833				
REL1	15.1132	4.641	.597	.810
REL2	15.3821	4.389	.657	.793
REL3	15.0425	4.903	.606	.808
REL4	15.3302	4.137	.702	.780
REL5	15.1509	4.726	.612	.806
Responsiveness Scale (RES): Cronbach's Alpha = 0.868				
RES1	11.8915	2.950	.681	.847
RES2	11.7547	3.456	.662	.857
RES3	11.6745	2.685	.763	.814
RES4	11.7736	2.830	.796	.798
Assurance Scale (ASS): Cronbach's Alpha = 0.861				
ASS1	11.9009	2.033	.678	.835
ASS2	11.7830	1.953	.708	.823
ASS3	11.8208	1.996	.643	.851
ASS4	11.8302	1.924	.814	.781
Empathy Scale (EMP): Cronbach's Alpha = 0.875				
EMP1	15.5991	3.199	.695	.850
EMP2	15.5047	2.962	.674	.858
EMP3	15.6840	3.080	.725	.843
EMP4	15.6557	3.099	.714	.845
EMP5	15.6132	3.129	.719	.844
Tangibles Scale (TAN): Cronbach's Alpha = 0.825				
TAN1	11.6698	2.469	.597	.802
TAN2	11.8632	2.394	.712	.756
TAN3	12.0472	2.225	.603	.806
TAN4	11.8255	2.164	.707	.752
Overall Service Quality Scale (OSQ): Cronbach's Alpha = 0.894				
OSQ1	11.9292	3.744	.758	.866
OSQ2	11.8632	3.503	.775	.859
OSQ3	11.9198	3.676	.745	.870
OSQ4	11.9481	3.490	.784	.856
OSQ5	11.9292	3.744	.758	.866
Customer Satisfaction Scale (SAT): Cronbach's Alpha = 0.811				
SAT1	12.0566	2.869	.598	.778
SAT2	12.1840	2.739	.642	.757
SAT3	12.2547	2.911	.624	.765
SAT4	11.9858	2.914	.654	.752

Source: Survey results of the Author (2017).

method and Varimax rotation method. The cumulative extraction variance is 63.933% (> 50%) and Eigenvalue 2.557 greater than 1 which are appropriate and meets the requirements. The factors loading of all 4 variables are greater than 0.5 so they are appropriate and meet the requirements. Therefore, the author will use these 4 observation variables with 1 factor for further analysis.

4.4. Results of Confirmed Factor Analysis (CFA)

4.4.1. Results of Composite Reliability and Average Variance Extracted of research concepts

The results in Table 10 show that all the concepts meet the requirements of Composite Reliability and Average Variance Extracted which Composite Reliability need 0.7 or above and Average Variance Extracted need 0.5 or above (Fornell and Larcker (1981), Hair, Black, Babin, and Anderson (2010)), the details of CFA analysis for each single factor of the research model will be presented in Appendix.

4.4.2. Results of Unidimesionality, Convergent Validity and Discriminant Validity of research concepts

The results shown in Fig. 7 show that the model has $Chi^2 = 467.572$; df = 384; $Chi^2/df = 1.218 < 5$, p-value = 0.002 (<0.05) was not appropriate due to the size of the sample (the author only surveyed 212 people); however, other fit indexes such as TLI = 0.969 > 0.9, CFI = 0.973 > 0.95, RMSEA = 0.032 < 0.07 are fit. Therefore, the saturated model is still compatible with market data.

The correlation coefficients and covariances associated with the standard errors (correlations coefficients and covariances value tests between factors in CFA analysis will represented in Appendix) all are differ from 1 (in other words, the concepts of research meet the requirements for Discriminant Validity (Hair et al., 2010). The error of the measurement variables is not correlated so the observational variables achieve the Unidimesionality (Steenkamp & Van Trijp, 1991), the standardized regression weights (λ_i) are greater than 0.5 and the regression weights are statistically significant, so the observational variables achieve the Convergent Validity (Gerbing & Anderson, 1988). In summary, the research model achieves the Unidimesionality, Convergent Validity and Discriminant Validity.

4.5. Results of modeland research hypotheses testing

4.5.1. Results of model testing

The Structural Equation Model (SEM) is processed by Amos Graphics software version 20.0 will be illustrated in the following Fig. 8.

Table 8
EFA results of quality of service and its components.

Kaiser–Meyer–Olkin measure of sampling adequacy					.832
Bartlett's Test of Sphericity	Approx. Chi-Square				2223.165
	df				231
	Sig.				.000
Variables	Component				
	1	2	3	4	5
EMP3	.825				
EMP5	.824				
EMP4	.821				
EMP2	.791				
EMP1	.768				
REL4		.812			
REL2		.779			
REL3		.760			
REL1		.730			
REL5		.728			
RES3			.865		
RES4			.853		
RES1			.805		
RES2			.791		
ASS4				.902	
ASS2				.835	
ASS1				.813	
ASS3				.780	
TAN4					.836
TAN2					.833
TAN1					.767
TAN3					.706
Kaiser–Meyer–Olkin measure of sampling adequacy					.834
Bartlett's test of sphericity	Approx. Chi-Square				486.715
	df				6
	Sig.				.000
Variables	**Component**				
	1				
OSQ4	.883				
OSQ2	.878				
OSQ1	.866				
OSQ3	.857				

Source: Survey results of the Author (2017).

Table 9
EFA results of customer satisfaction.

Kaiser–Meyer–Olkin measure of sampling adequacy		.755
Bartlett's test of sphericity	Approx. Chi-Square	277.881
	df	6
	Sig.	.000
Variables	Component	
	1	
SAT4	.818	
SAT2	.807	
SAT3	.798	
SAT1	.775	

Source: Survey results of the Author (2017).

The results shown in Fig. 8 show that the model has $Chi^2 = 545.827$; df = 399; $Chi^2/df = 1.368 < 5$, p-value = 0.000 (<0.05) was not appropriate due to the size of the sample (the author only surveyed 212 people); however, other fit indexes such as TLI = 0.948 > 0.9, CFI = 0.952 > 0.95, RMSEA = 0.042 < 0.07 are fit. Therefore, this model is still consistent with data collected from the market.

4.5.2. Results of research hypotheses testing

We will test all the research hypotheses at 5% significant level; level of confidence is 95%. Table 11 will show the results of hypotheses testing.

Table 10
Results of composite reliability and average variance extracted of research concepts.

Research concepts	Code	Number of observational variables	Composite reliability (ρ_C)	Average variance extracted (ρ_{VC})
Reliability	REL	5	0.834	0.503
Responsiveness	RES	4	0.873	0.633
Assurance	ASS	4	0.866	0.621
Empathy	EMP	5	0.876	0.586
Tangibles	TAN	4	0.832	0.554
Overall service quality	OSQ	4	0.894	0.678
Customer satisfaction	SAT	4	0.812	0.520

Source: Survey results of the author (2017).

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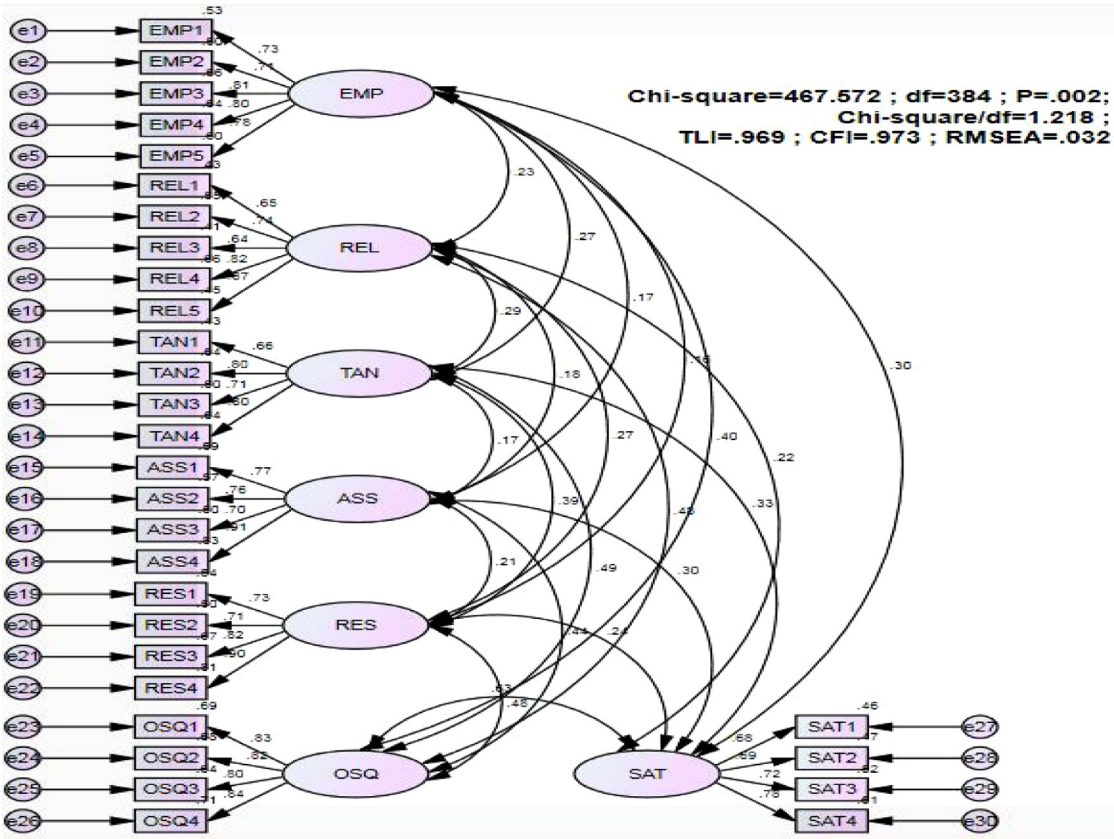

Fig. 7. CFA analysis for research model

Source: Survey results of the Author, 2017.

Table 11
Results of research hypotheses testing.

The relationship			Estimate	S.E.	C.R.	*P*	Results
OSQ	←	EMP	.342	.093	3.659	***	Accept H_1
OSQ	←	REL	.340	.082	4.143	***	Accept H_2
OSQ	←	ASS	.390	.083	4.679	***	Accept H_3
OSQ	←	RES	.267	.066	4.063	***	Accept H_4
OSQ	←	TAN	.351	.092	3.817	***	Accept H_5
SAT	←	OSQ	.530	.084	6.310	***	Accept H_6

Source: Survey results of the author (2017).

The results presented in Table 11 show that p-value of all tests are lower than 0.05 so all research hypotheses are accepted at 5% significant level and a 95% level of confidence

4.5.3. Discussion the data analysis results

First of all, the quality of logistics services at Cat Lai Port is directly affected by five components: reliability, responsiveness, assurance, empathy and tangibles with at 5% significant level and 95% level of confidence. (Hypotheses H_1, H_2, H_3, H_4, H_5 are accepted). All estimates value of these five variables are marked (+), meaning that these factors are positively correlated with the OSQ factor and it is suitable with the hypotheses in the proposed research model. This means that customers using logistics services at Cat Lai Port will highly appreciate the quality of service when they can rely on the services provided by Cat Lai Port; when they think that Cat Lai Port has good facilities, tangible means; when they highly appreciate the assurance of Cat Lai Port; when their expectations from Cat Lai Port are responsive and when they feel Cat Lai Port always shows empathy to them.

Moreover, customer satisfaction about logistics services at Cat Lai Port is directly affected by the overall service quality provided by Cat Lai Port with 5% significant level and 95% level of confidence (H6 hypothesis is accepted). Estimate value of this relationship is marked (+), meaning that OSQ factor are positively correlated with the SAT factor and it is suitable with the hypotheses in the proposed research model. This means that when customers appreciate the quality of logistics services that Cat Lai Port provided, it will make them feel satisfied.

5. Conclusion and implications

To sum up what has been empirically explored, port logistics service quality is confirmed to be influenced by five components of SERVQUAL model. Specifically, assurance, tangibles and empathy are the most influential factors of port logistics service quality, while the following others are reliability and responsiveness. This means that when using port logistics services, customers concern most about strong commitment of the port reflected by the staff professionalism in dealing with customers' problems,

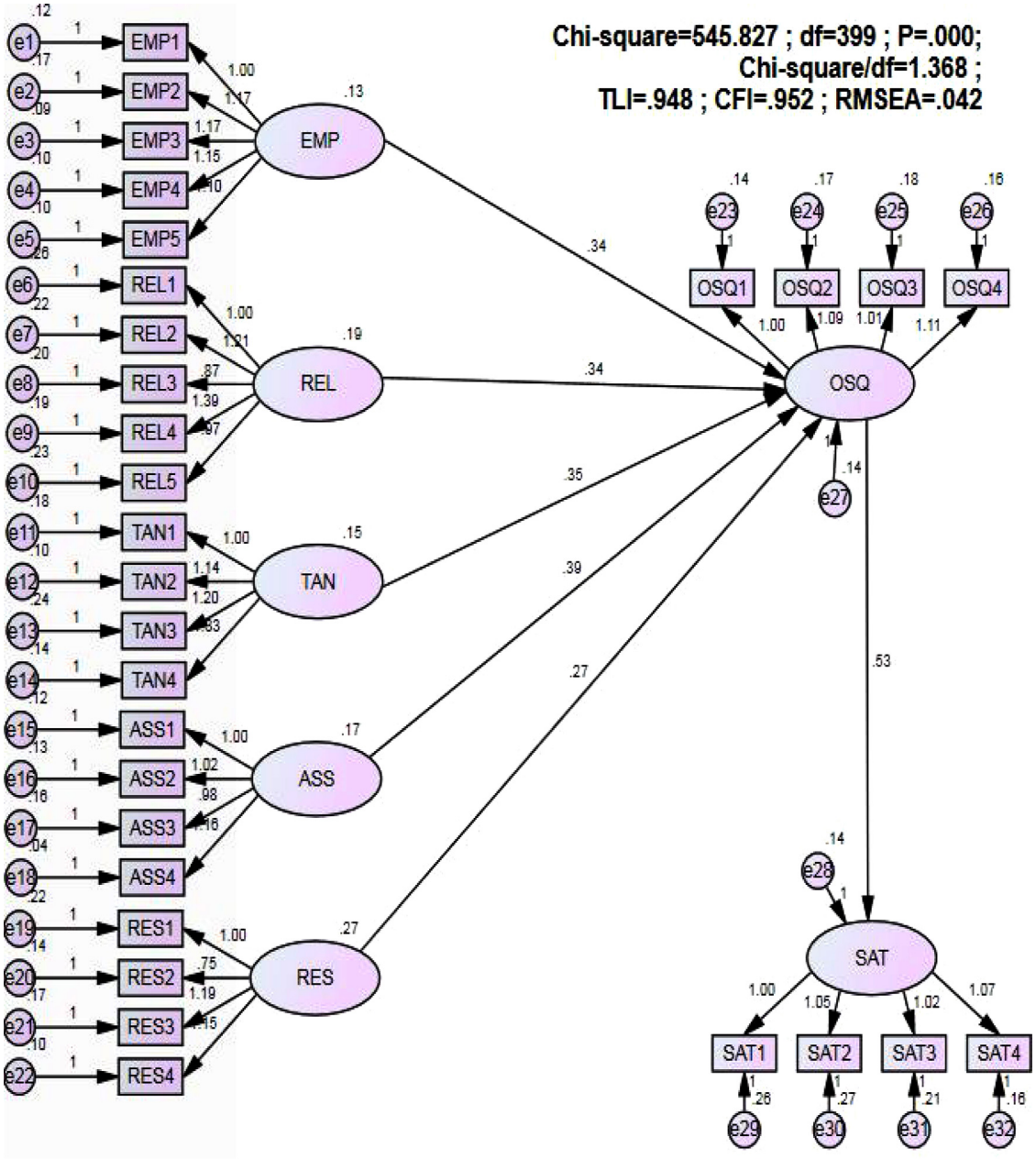

Fig. 8. SEM research model.

Source: Survey results of the Author, 2017.

particularly the safety of their shipments, punctual clearance and on-time delivery or receipt of consignments. In addition, the valid effect of tangibles reveals the roles of technology and infrastructural investment in increasing service quality which ultimately enhances customer satisfaction. Meanwhile, the degree to which port staff put themselves into their customer's situations determines whether port logistics services quality is sufficient. The two remaining less influential factors (reliability and responsiveness) are also important to customers but they are not their priorities. In addition, customer satisfaction appears to be positively affected by overall service quality which suggests the strategic focus to gain market share in port logistics services industry. Given all hypotheses are supported, it could be concluded that *the road to customer satisfaction is paved with good service quality* and a five-fold strategy needs to be prepared by port management prior to *entering that road*. *From the theoretical perspective*, this study contributes to the contemporary literature on service quality in the context of port logistics services industry in developing and transitional economies. The results of the study suggest that port logistics services quality could be multi-dimensionally measured, which could provide practitioners with clearer implications for which operational activities need to be intervened by port management. Findings are also the starting point for formulating managerial solutions to compete effectively in the age of port privatization in historically centrally-planned economies like Vietnam which is characterized by the dominance of state ownership in certain sectors including port logistics services. *From the managerial perspective*, practitioners may find the following proposed solutions helpful in enhancing port logistics services quality thereby increased customer satisfaction.

Regarding the *assurance-port logistics services quality* link, port management should raise staff awareness of *customer-oriented culture* and equip their staff with essential skills and proper attitudes. Forming *a uniform code of conduct* in service provisioning which

is the basement for monthly and annual performance assessment could be solution to spread the culture through the institution. In addition, port management could design a policy of recruiting experienced employees to ensure the most effective quality of work in order to quickly capture customer requirements and come up with effective resolutions. To do so, port management could develop intensive short-term training courses to improve staff competency and expertise relevant to the industry.

Regarding the *tangibles-port logistics services quality* link, firstly, port management should focus on enhancing and expanding current port infrastructural items and dredging canals to avoid waterway and intra-port traffic jams and increase transport capacity. Secondly, warehousing facilities need to be expanded because it is one of the cutting edge of ports in logistics services provision as most firms in developing countries are not capable of investing in large-scale and costly items. In addition, port management should regularly purchase modern and high-performing specialized loading and unloading equipment from prestigious suppliers. Simultaneously, it is necessary to replace outdated machines and equipment which are prone to malfunctioning even if they have been regularly repaired, which ensures the workflow is smooth and uninterrupted. Besides, since working office is the item where administrative procedures take place, decoration and professional design are essential to bring about good experience for field staff of client firms when interacting with port staff and using port logistics services.

Regarding the *empathy-port logistics services quality* link, firstly, port management should design training programs to improve interpersonal communication competence, problem-solving skills and working attitudes of the workforce. Especially, strengthening foreign languages literacy for employees is an important task as the field of international logistics requires a large number of terminologies. Secondly, port management should create a platform and a hotline for customers to give their feedbacks about port logistics services as it is easier for businesses to listen to their customers which enhance the perceived staff dedication of customers. Thirdly, rewarding policy needs to be commensurate with staff performance and dedication which nurtures dynamic and positive mindset of port staff when providing services to customers especially in difficult and urgent cases.

Regarding the *reliability-port logistics services quality* link, firstly, port management should apply advanced management practices to all intra-port areas including piers, docks, yards and warehouses, which significantly improves the efficiency in port operations. Secondly, port management should continuously streamline the procedures of customs clearance and processes of goods delivery/receipt to save time and costs for customers. Thirdly, port management should allow more shipping lines to operate in the port because it will create favorable conditions for customers to use the port logistics services as they have a wider range of choices of shipping lines.

Regarding the *responsiveness-port logistics services quality* link, firstly, port management should focus on making more services available to customers by expanding value-added services such as picking up, packing, managing and controlling inventory, bar code labeling and door-to-door delivery toward at competitive charges and fees. It will provide the all in one choice for customers when they consider the port to handle their shipments. Secondly, port management should strengthen collaborative ties with international maritime organizations, research institutes and universities around the world and also appoint qualified and talented young managers to undertake long-term and short-term training courses abroad to learn advanced port management experiences and methods in advanced industrial countries. Thirdly, port management should have policies for proper resolutions of complaints and indemnities. In the shipping logistics, it sometimes happen force majeure or loss of cargoes that nobody want it to occur. Therefore, these policies help the port be able to resolve problems quickly and provide the most appropriate support to customers, thereby enhancing the prestige of the port as well as increasing customer satisfaction. Finally, port management should develop its own smartphone application software. This application will make the process of creating and submitting cargoes documents become more quickly, accurately and systematically, thereby minimizing mistakes, reducing management and personnel costs and saving time for port operations. If the entire operational process of cargo handling process is digitalized, port staff could be able to avoid information loss and look up any necessary information more rapidly, especially when there is a sudden problem. This will enhance the quality of logistics services and global competitiveness by helping port staff respond quickly to customers' requirements.

To some extent, future research is potential in the following directions. Firstly, further studies should be carried out with larger samples to increase the coverage of the sample and represents the research majority. In addition, extending the scope of geographic coverage to more ports is also necessary. Secondly, the research object is the port logistics services at Cat Lai Port in general, not concentrating on a specific type of port services such as inspection, warehousing, clearance of import documents, etc. Future studies therefore need to research on specific types of port logistics services. Finally, the proposed research model is based on the five-component service quality model (SERVQUAL) of Parasuraman et al. (1988) to evaluate quality of logistics services and customer satisfaction. However, in fact, port logistics services quality is also influenced by other factors such as services prices, geographical location and port area, procedures of processing goods or quality of relationship with customers, Therefore, future studies could consider the impacts of those factors to provide more comprehensive solutions to improve quality of port services and satisfy customers.

Conflict of interest

None.

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Appendix 1.

List of respondents of focus group interview

No.	Name	Position	Email
1	Phan The Hung	Field staff	hungphanlogistics@gmail.com
2	Nguyen Van Lam	Field staff	lamnguyenforwarding@gmail.com
3	Tran Trong Van	Field staff	trongvan0112@gmail.com
4	Hoang Kim Long	Field staff	hoanglongservice@gmail.com
5	Nguyen Huu Hoang	Field staff	hoangnguyenfreight@gmail.com
6	Van Kim Hung	Field staff	vankimhung2112@gmail.com
7	Nguyen Manh Truong	Field staff	truongnguyentrading@gmail.com
8	Hoang Van Minh	Field staff	hoangvanminh131279@gmail.com
9	Pham The Vinh	Field staff	vinhpham77777@gmail.com
10	Mai Van Nam	Field staff	maivannamforwarder@gmail.com

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